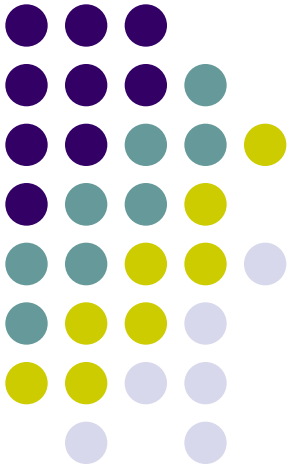
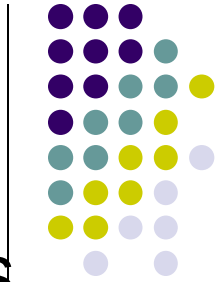
ENGINEERING ECONOMY

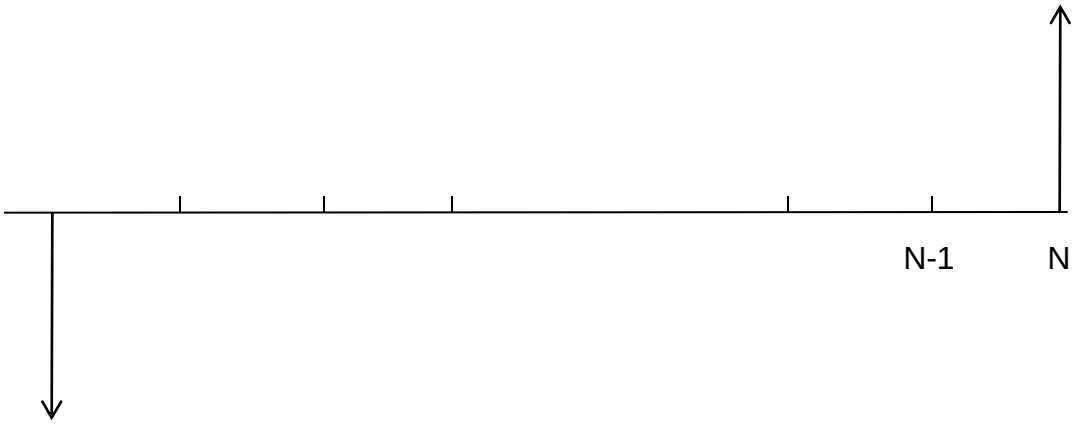
Factors and their Use



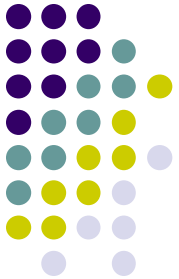
The concept of equivalence



- Alternatives should be compared as far as possible when they produce similar results, serve the same purpose, or accomplish the same function.
- This is not always possible in some types of economy studies.
- The question is how can alternatives for providing the same service or accomplishing the same function be compared when interest is involved over an extended period of time?

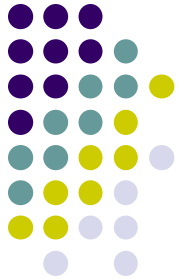

N-1
N

Single-Payment Factors (F/P and P/F)



- *If we have P cedis at the present time and invest it at an interest rate of i, the future value F in 1 year will be*
  - $F_1 = P + Pi$
  - $F_1 = P\ (1 + i)$
- After two years it will be
  - $F_2 = F_1 + F_1 i$
  - $= P\ (1 + i) + P\ (1 + i)i$
  - $= P\ (1 + i)^2$

F/P factor



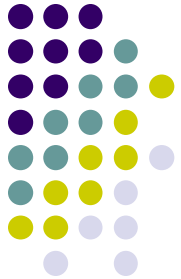
- Similarly, the amount of money accumulated at the end of year three, using will be

$$F_3 = F_2 + F_2 i$$

- Substituting P $(1 + i)^2$ for F_2 and simplifying,

$$F_3 = P(1+i)^3$$

F/P factor



- From the preceding values, it is evident by mathematical induction that the formula can be generalized for n years to

$$F = P\left(1+i\right)^{n}$$

F/P factor

- The factor

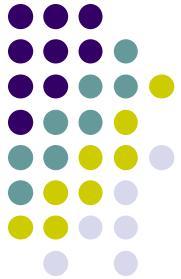
$$(1 + i)^n$$

is called

the single payment compound amount factor (SPCAF)

but it is usually referred to as F/P factor.

P/F factor



- Solving for P in the last equation in terms of F results in the expression

$$P = F\left[\frac{1}{(1+i)^n}\right]$$

P/F factor

- The expression in the brackets is known as the

  single payment present-worth factor (SPPWF),

  or the **P/F factor**

$$P = F\left[\frac{1}{(1+i)^n}\right]$$

Uniform series

- The present worth P of a uniform series can be determined by considering each A as a future worth F and using the equation with the P/F factor and then summing the present-worth values. The general formula is

$$P = F\left[\frac{1}{(1+i)^n}\right]$$

$$P = A\left[\frac{1}{(1+i)^1}\right] + A\left[\frac{1}{(1+i)^2}\right] + A\left[\frac{1}{(1+i)^3}\right] + \ldots A\left[\frac{1}{(1+i)^{n-1}}\right] + A\left[\frac{1}{(1+i)^n}\right]$$

Uniform series

- where the terms in brackets represent the P/F factors for years 1 through n respectively. Factoring out A, (eq.4)

$$P = A\left[\frac{1}{(1+i)^1} + \frac{1}{(1+i)^2} + \frac{1}{(1+i)^3} + \ldots + \frac{1}{(1+i)^{n-1}} + \frac{1}{(1+i)^n}\right]$$

Uniform series

- The equation 4 may be simplified by multiplying both sides of the equation by the (P/F,*i*%,1) factor which is 1/(1 + *i*)

$$\frac{P}{1+i} = A\left[\frac{1}{(1+i)^2} + \frac{1}{(1+i)^3} + \frac{1}{(1+i)^4} + \ldots + \frac{1}{(1+i)^n} + \frac{1}{(1+i)^{n+1}}\right]$$

(Eq. 5)

Uniform series

- Subtracting equation 4 from equation 5, simplifying, to obtain the expression for P when i ≠ 0

$$\frac{P}{1+i} = A\left[\frac{1}{(1+i)^2} + \frac{1}{(1+i)^3} + \frac{1}{(1+i)^4} + \ldots + \frac{1}{(1+i)^n} + \frac{1}{(1+i)^{n+1}}\right]$$

and then dividing both sides of the relation by -*i*/(1 + *i*) leads to an expression for P when *i* ≠ 0

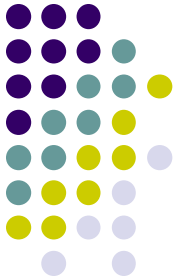
$$P = A\left[\frac{(1+i)^n - 1}{i(1+i)^n}\right]$$

Uniform-series

- The term in brackets is called the **Uniform-series present-worth factor (USPWF),** or P/A factor.
- This equation will give the present worth P of an equivalent uniform annual series A which begins at the end of year 1 and extends for n years at an interest rate i.

$$P = A\left[\frac{(1+i)^n - 1}{i(1+i)^n}\right]$$

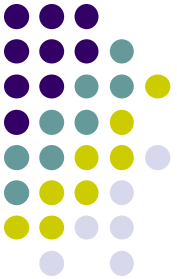
Uniform Series



- Rearranging the P/A factor we get the Capital Recovery Factor (CRF) or the A/P factor
- This yields the equivalent uniform annual worth A over n years of a given investment P when the interest rate is i.

$$A = P\left[\frac{i(1+i)^n}{(1+i)^n - 1}\right]$$

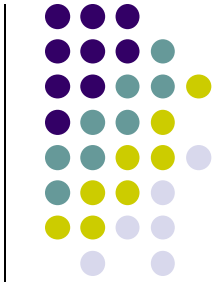
P/A and A/P FACTORS



- ***It is very important to remember that these formulas are derived with the present worth P and the first uniform annual amount (A) one year (period) apart.***

- ***That is the present worth P must always be located one period prior to the first A.***

Sinking fund factor and Uniform-series compound-amount factor (A/F and F/A)

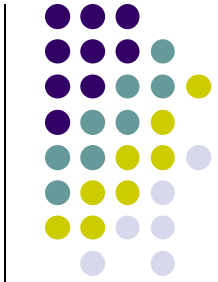


- The simplest way to derive the formulas is to substitute into those already developed. Thus, if P from equation for P/F is substituted into equation for A/P, the following formula results:

$$A = F\,\frac{1}{(1+i)^n}\,\frac{i(1+i)^n}{(1+i)^n - 1}$$

$$= F\left[\frac{i}{(1+i)^n - 1}\right]$$

Uniform series



- The expression in brackets is the sinking fund, or A/F, factor.
- It is use to determine the uniform annual worth series that would be equivalent to a given future worth F.
- ***Note that the uniform series A begins at the end of period 1 and continues through the period of the given F.***

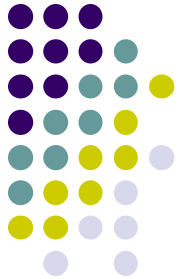
Uniform Series

- Rearranging to express F in terms of A, we get the equation below. The term in bracket is the

  Uniform Series Compound Amount Factor (USCAF) or **F/A factor**

$$F = A\left[\frac{(1+i)^n - 1}{i}\right]$$

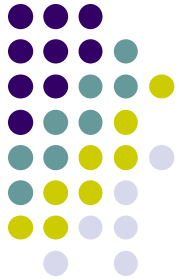
F/A factor



- The uniform series compound amount factor (USCAF), or F/A factor when multiplied by a given uniform annual amount A, will yield the future worth of the uniform series.

- ***It is important to note that the future amount F occurs in the same period as the last A.***

Standard Factor notation and the use of interest tables



- A standard notation has been adopted which includes the interest rate and the number of periods and is always in the general form:

- (*X*/*Y*, *i*, *n*)

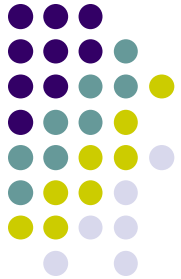
  X represents what is to be found

  Y represents what is given

  i is the interest rate in percent

  n is the number of periods involved

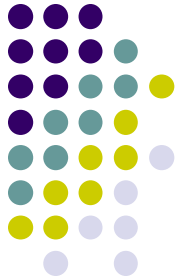
Standard Factor notation and the use of interest tables



- (F/P, 6%, 20)

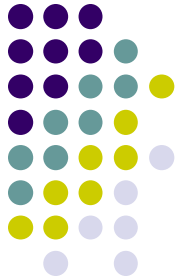
  means obtain the factor which when multiplied by a given P allows you to find the future amount of money F that will be accumulated in 20 periods if the interest rate is 6% per period.

Standard Factor notation and the use of interest tables



- Factor Name and Standard Notation
- Single payment present-worth (SPPWF) or the P/F (*P/F, i, n*)
- Single payment compound amount (SPCAF) or the F/P (*F/P, i, n*)
- Uniform-series present-worth (USPWF), or P/A.(*P/A, i, n*)
- Capital recovery factor (CRF), or the A/P(*A/P, i, n*)
- Sinking fund, or A/F(*A/F, i, n*)
- Uniform series compound amount (USCAF), or F/A(*F/A, i, n*)

Interpolation in interest tables



- Sometimes it is necessary to locate a factor value for an interest rate *i* or number of periods *n* that is not in the interest tables. When this situation occurs, the desired factor value can be obtained in one of two ways:

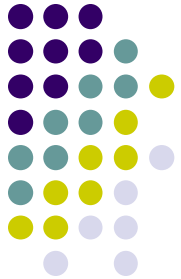
- By using the formulas derived or
- By interpolating between the tabulated values

Interpolation in interest tables

- When the formula is applied, the factor value is accurate since the specific values of I and n are used. However it is possible to make mistakes since the formulas are similar to each other, especially with uniform series.

- Additionally, the formulas become more complex when gradients are introduced.

Interpolation in interest tables



- Linear interpolation for an untabulated interest rate I or number of periods n takes more time to complete than using the formula.
- Also interpolation introduces some level of inaccuracy depending on the distance between the two boundary values selected for i or n, as the formulas themselves are non-linear functions.